

EGR 345 HW 2: Transfer Functions

Find the transfer functions for the following differential equations:

Note: Assume that x and y are output variables and u and f are inputs.

1. $\dot{x} + 7x(t) = u(t)$

2. $3\dot{y} + 5y(t) = u(t)$

3. $a\dot{y} + by(t) = cu(t)$, assuming a , b , and c are constants

4. $4\dot{x} = 8f(t)$

5. $3\ddot{x} + 4\dot{x} + 25x(t) = f(t)$

6. $\ddot{x} + 2\dot{x} + 49x(t) = 3f(t)$

7. $m\ddot{x} + b\dot{x} + kx(t) = f(t)$, assuming m , b , and k are constants

8. $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x(t) = \frac{1}{m}f(t)$, assuming ζ , ω_n , and m are constants